

Malware Analysis Report For MachO-OSX-x64-ls

Binary Type

Mach-O (Mach Object). (Approved)

Analyzing File

- **File Name:** MachO-OSX-x64-ls
 - **File Extension:**
 - **File Type:** Mach-O (Executable)
 - **MD5 Hash:** d174dcfb35c14d5fcaa086d2c864ae61
 - **SHA256 Hash:** bb38e04ca01881df5e6b92e2231f3173ee6d610b32af3068e8fe6b001c51a10f
-

File Identification

File detected as a valid macOS/iOS executable.

Mach-O File Analysis

Mach-O File Analysis: True

Mach-O File Imports

- libSystem.B.dylib
- libncurses.5.4.dylib
- libutil.dylib

Mach-O File Exports

- `radr://5614542`
-

String Analysis

ASCII Strings

- `__PAGEZERO`
- `__TEXT`
- `__text`
- `__TEXT`
- `__stubs`
- `__TEXT`
- `__stubhelper`
- `__TEXT`
- `__cstring`
- `__TEXT`
- `__unwindinfo`
- `__TEXT`

- `_ehframe`
- `__TEXT`
- `__DATA`
- `_programvars`
- `__DATA`
- `__DATA`
- `_nlsymbol_ptr`
- `__DATA`
- `_lasybol_ptr`
- `__DATA`
- `__data`
- `__DATA`
- `__const`
- `__DATA`
- `__common`
- `__DATA`
- `__DATA`
- `__LINKEDIT`
- `/usr/lib/dyld`
- `/usr/lib/libncurses.5.4.dylib`
- `/usr/lib/libutil.dylib`
- `/usr/lib/libSystem.B.dylib`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `AWAVAUATSH`
- `([A\A]A^A_`
- `AWAVATSH`
- `[A\A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `8[A\A]A^A_`
- `8[A\A]A^A_`
- `AWAVAUATSH`
-
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `AWAVAUATSH`
- `[A\A]A^A_`
- `$FreeBSD: src/bin/lscmp.c,v 1.12 2002/06/30 05:13:54 obrien Exp $`
- `LS_COLWIDTHS`
- `malloc`

- %llu : %qu : %lu : %i : %i : %i : %qu : %lu : %lu
- %s: %s
- fflagstotr
- fts_open
- %s: directory causes a cycle
- bin/ls
- fts_read
- COLUMNS
- CLICOLOR
- 1@ABCFGHLOPRSTUWabcdefghijklmnopqrstuvwxyz
- CLICOLOR_FORCE
- LSCOLORS
- \$FreeBSD: src/bin/ls/ls.c,v 1.66 2002/09/21 01:28:36 wollman Exp \$
- exfxcxdxbxegedabagacad
- warn: LSCOLORS should use characters a-h instead of 0-9 (see the manual page)
- error: invalid character '%c' in LSCOLORS env var
- %*llu
- Unix2003
- total %qu
- %e %b %T %Y
- %b %e %T %Y
- %e %b %R
- %b %e %R
- %e %b %Y
- %b %e %Y
- ls: %s: %s
- Unable to translate qualifier on ACL
- unknown
- inherited
- %d: %s%s %s
- add_file
- execute
- search
- delete
- append
- add_subdirectory
- delete_child
- readattr
- writeattr
- readextattr
- writeextattr
- readsecurity
- writesecurity
- file_inherit
- directory_inherit
- limit_inherit
- only_inherit
- %s%s %*u
- %s%s %u %-s
- %s%s %u %-s %-*s
- %3d, 0x%08x
- %3d, %3d

- %s%qu
- \$FreeBSD: src/bin/ls/print.c,v 1.57 2002/08/29 14:29:09 keramida Exp \$
- usage: ls [-ABCFGHLOPRSTUWabcdefghiklmnopqrstuw1] [file ...]
- \$FreeBSD: src/bin/ls/util.c,v 1.38 2005/06/03 11:05:58 dd Exp \$
- UG`JBp
- @__DefaultRuneLocale
- @_stackchkguard
- @_stderr
- @_stdout
- @_optind
- @dyldstubbinder
- @_error
- @_maskrune
- @_snprintf_chk
- @_stackchkfail
- @_tolower
- @aclfree
- @aclget_entry
- @aclgetflagnp
- @aclgetflagsetnp
- @aclgetlinknp
- @aclgetpermnp
- @aclget_permset
- @aclget_qualifier
- @aclgettagtype
- @_atoi
- @compatmode
- @_exit
- @_fflagstostr
- @_fprintf
- @_fputs
- @_free
- @ftschildren\$INODE64
- @ftsclose\$INODE64
- @ftsopen\$INODE64
- @ftsread\$INODE64
- @ftsset\$INODE64
- @_getbsize
- @_getenv
- @_getgrgid
- @_getopt
- @_getpid
- @_getpwuid
- @_getuid
- @_getxattr
- @groupfrom_gid
- @humanizenumber
- @_ioctl
- @_isatty
- @_kill
- @_listxattr

- `@_localtime`
- `@_malloc`
- `@mbuuidtoid`
- `@mbuuidtostring`
- `@_mbrtowc`
- `@_memchr`
- `@nl_langinfo`
- `@_printf`
- `@_putchar`
- `@_readlink`
- `@_realloc`
- `@_setenv`
- `@_setlocale`
- `@_signal`
- `@_snprintf`
- `@_sscanf`
- `@_strcoll`
- `@_strcpy`
- `@_strdup`
- `@_strerror`
- `@_strftime`
- `@_strlen`
- `@_strmode`
- `@_tgetent`
- `@_tgetstr`
- `@_tgoto`
- `@_time`
- `@_tputs`
- `@userfrom_uid`
- `@_warn`
- `@_warnx`
- `@_wcurwidth`
- `@_write`
- `_mhexecute_header`
- `UCUUUL:LLLn`
- `_mhexecute_header`
- `__DefaultRuneLocale`
- `_error`
- `_maskrune`
- `_snprintf_chk`
- `_stackchkfail`
- `_stackchkguard`
- `_stderrp`
- `_stdoutp`
- `_tolower`
- `acfree`
- `ac/get_entry`
- `ac/getflagnp`
- `ac/getflagsetnp`
- `ac/getlinknp`
- `ac/getpermpnp`

- *ac/get_permset*
- *ac/get_qualifier*
- *ac/gettagtype*
- *compatmode*
- *_fflagstostr*
- *_fprintf*
- *_fputs*
- *ftschildren\$INODE64*
- *ftsclose\$INODE64*
- *ftsopen\$INODE64*
- *ftsread\$INODE64*
- *ftsset\$INODE64*
- *_getbsize*
- *_getenv*
- *_getgrgid*
- *_getopt*
- *_getpid*
- *_getpwuid*
- *_getuid*
- *_getxattr*
- *groupfrom_gid*
- *humanizenumber*
- *_ioctl*
- *_isatty*
- *_listxattr*
- *_localtime*
- *_malloc*
- *mbnuuidtoid*
- *mbnuuidtostring*
- *_mbrtowc*
- *_memchr*
- *nllanginfo*
- *_optind*
- *_printf*
- *_putchar*
- *_readlink*
- *_realloc*
- *_setenv*
- *_setlocale*
- *_signal*
- *_snprintf*
- *_sscanf*
- *_strcoll*
- *_strcpy*
- *_strdup*
- *_sterror*
- *_strftime*
- *_strlen*
- *_strmode*
- *_tgetent*
- *_tgetstr*

- _tgoto
- _tputs
- userfrom_uid
- _warnx
- _wcwidth
- _write
- dyldstubfinder
- radr://5614542
- com.apple.ls
- libncurses.5
- com.apple.libutil1
- libSystem.B
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 070214211919Z
- 150214211919Z0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- drmljGa
- %http://www.apple.com/appleca/root.crl0
- QKK:laN
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- 060425214036Z
- 350209214036Z0b1
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- https://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- 070223220256Z
- 150114220256Z0V1
- Apple Inc.1
- Apple Software1
- Software Signing0
- EA3%|k=
- http://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- ,http://www.apple.com/appleca/codesigning.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority

UTF-8 Strings

No detected.

UTF-16LE Strings

No detected.

Extracted URLs

- <http://www.apple.com/appleca/root.crl0>
- <https://www.apple.com/appleca/0>
- <http://www.apple.com/appleca/0>
- <http://www.apple.com/appleca/codesigning.crl0>

Base64 Strings

- Possible Base64 Encoded String: com/appleca/root -> Decoded: rjey-
- Possible Base64 Encoded String: 150114220256Z0V1 -> Decoded: 5nzgEu

File Entropy Analysis

- **Entropy:** 5.02
- **Status:** This file likely contains Standard Text or human-readable text.

VirusTotal Analysis

Scan Date: 2024-07-01 14:50:32

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 66
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 12

Community Votes

- Harmless: 0
- Malicious: 0

Virus Total Permalink

- No VirusTotal link available.

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results